

Exploring Transitional Automation with New and Old Drivers

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Abstract

Age and experience influence driver ability to cope with transitions between automated and manual driving, especially when drivers are engaged in media use. This study evaluated three age cohorts (young/new drivers, adults, and seniors) on their performance in transitions from automated driving to manual vehicle control in a laboratory driving simulator. Drivers were given three tasks to perform during the automated driving segments: to watch a movie on a tablet, to read a story on a tablet, or to supervise the car's driving. We did not find significant differences in people's driving performance following the different tasks. We also did not find significant differences in driving performance between the people in each age group who successfully completed the study; however, the rejection rate of the senior age group was over 30% because many of the people in this age group had difficulty hearing instructions, understanding tasks, or remembering what to do. We also observed pronounced differences in typical driving behaviors between groups which designers of automated systems need to take into account.

Introduction

Most near-future visions of automated driving will require the driver to take control of the vehicle when road conditions present situations beyond the capabilities of the automated system. This driving stage (NHTSA 2-3) ("U.S. Department of Transportation Releases Policy on Automated Vehicle Development | National Highway Traffic Safety Administration (NHTSA)" 2014) poses serious design challenges for automotive human-machine interface designers, primarily regarding signaling transition events when the driver must resume driving, and also around creating situation awareness so that the driver is well prepared to drive after resuming control. This study explored the effect of age on ability to control a simulated vehicle in structured handoffs from an automated driving system, and to navigate potential accident situations immediately post-transition without automation support.

New drivers and elderly drivers have different risk profiles, different concerns, and different needs which must be addressed in order for automated partially-automated driving to increase safety and comfort. This diversity may present special challenges for the design of automated driving systems, and this study explored the behavior of these populations in structured transitions from automated driving to driver control in hazardous driving situations.

Prior Research

Older Drivers

Automated driving technology has been touted as a major benefit, particularly for aging drivers, (e.g. (Vercruyssen 1998)) because drivers become less capable of driving as they age. Analysis of the Fatality Analysis Reporting System (FARS) and General Estimates System (GES) data from 2002-2006 indicates that while drivers 60-69 had collision rates similar to those of middle-aged drivers, drivers over 70 years of age had elevated risk levels in a variety of conditions, and drivers over 80 were overrepresented in crashes (Stutts, Martell, and Staplin 2009). However, these very challenges that make older drivers good candidates for assistance from automated vehicle technologies also make human-machine interaction design for these drivers particularly difficult. Older drivers are more likely to have difficulty seeing (Eby et al. 1998) or managing complex driving environments, (Ball and Owsley 1991) but they are also more conservative drivers, limiting exposure to challenging situations (Hakamies-Blomqvist 1994). High frequency hearing loss, and difficulty filtering audio streams is common among the elderly (Davidse 2006; Dalton et al. 2003) and speech-based interfaces are more likely to confuse and distract older drivers (Lee et al. 2001). Loss of physical strength and range of motion due to aging reduces driver ability to control the vehicle, limiting peak brake force and steering ability (Davidse 2006). Base reaction times do not appear to increase with age (Lerner 1993), in contrast to most other faculties.

Young and New Drivers

Newly licensed drivers are about eight times more likely to be involved in a fatal crash than more experienced drivers (National Highway Traffic Safety Administration, National Center for Statistics and Analysis, and U.S. Department of Transportation 2012). Young drivers' traits contribute greatly to making driving the leading cause of death for people between the ages of 4 and 34 (Subramanian 2006; Lansdown 2002). Research on new drivers show that over the first few years of driving, the effects of experience greatly exceed those of age (McKnight and McKnight 2003). Newer drivers are relatively more likely to suffer from inattention, drive at high speed, fail to recognize hazards, and are less likely to execute necessary emergency procedures. Younger drivers are particularly susceptible to the lures of interacting with technology on the road, which is problematic because they have greater need to attend to the operational tasks of driving (Lansdown 2002). As a result of these differences from more experienced drivers, it is important to characterize the abilities of

novices in emergency situations shortly after a transition to manual driving after a period of automated driving.

Attention and Arousal Management

Automated driving allows for attention to be diverted from continuous assessment of the environment, reducing situation awareness and potentially impairing ability to regain control of the vehicle if required to do so (Endsley and Kiris 1995; Mkrtchyan et al. 2012). This reduction in cognitive workload will allow the sleep drive to exert greater influence over drivers who are disengaged, possibly increasing the time required to re-engage in driving and to react to hazardous situations. Driver drowsiness is a major hazard in driving—up to 20% of road collisions have drowsiness or sleepiness as a contributing factor (MacLean, Davies, and Thiele 2003); and approximately 3% are directly attributable to drowsiness or sleepiness (National Highway Traffic Safety Administration, National Center for Statistics and Analysis, and U.S. Department of Transportation 2012). Automated driving will eliminate some of these collisions, as automated systems will be able to drive in areas of low stimulation, and can respond to some road hazards independently.

Engagement in stimulating activities, such as visual media use currently considered to be hazardous during driving, may be desirable to keep drivers alert, and thus more able to take control of the vehicle than they would be if drowsy or asleep (Neubauer, Matthews, and Saxby 2014). Considering the desire to alleviate boredom and to engage in media use, engagement with media or communication technology while the vehicle is under automated control may also help to decrease collisions rates. This interaction must be limited to periods where the vehicle is under automated control, and prevented when the driver must be engaged in driving, as the relative risk from visual distraction is 3-5 times that of baseline driving and is a factor in 22% of vehicle collisions (2006).

Visual Media Interaction

Additionally, we are interested in whether engagement in different media interactions during automated driving would impede a driver's ability to regain control of the vehicle, and if they were at greater risk for crashes in the post-transition period. Distraction research in simulators around traditional manual driving suggests that driver performance is degraded by secondary tasks like cell phone usage (Stein, Parseghian, and Allen 1987; Zwahlen, Adams, and Schwartz 1988). However, studies looking at naturalistic driving data do not reflect increased risk of people using cell phones (Fitch et al. 2013). Studies looking at people interacting with telematics systems and manual tasks indicate that people compensate for increased complexity by increasing following distances (Greenberg et al. 2003). As the visual and auditory processing capabilities are semi-independent, auditory media or speech does not interfere with attention to the same degree as does visual interaction.

The deliberate introduction of non-driving activities (such as visual media consumption) into the driving environment represents a major break from prior research, which focused on the hazards of distracted driving. The National Safety Council (2012) estimates that 21% of automobile crashes in the US involve the use of a mobile phone, and that an additional 3% involved text messaging. The common assumption is that the increased prevalence of personal electronics and the wider array of activities offered on these devices will cause the problem of driving distraction to worsen in the years ahead. Counterintuitively, overall collision rates are decreasing, even as

personal electronics proliferate, and in-car use is increasing (Farmer, Braitman, and Lund 2010).

Research Questions for Partially-Automated Driving

Supervising an Advanced Driver Assistance System (ADAS) driving the vehicle under fully automated control, will leave the driver prone to drowsiness or inattention as a result of disengagement (Thiffault and Bergeron 2003) which could present a serious threat to safety if a driver who is not fully awake or is distracted must quickly take control of the vehicle. Disengagement from driving will present new challenges: being engaged in an alternate activity may forestall the drowsiness, (Neubauer, Matthews, and Saxby 2014) but then the challenge will be changing focus from media interaction to driving, and potentially managing a handheld device.

If drivers are to engage in alternate activities, switching from engagement in visual media to engagement in driving will have to be effected in a short time. For that to happen, a structured transition where the driver has time in which to shift attention resources, establish situation awareness, and to put away handheld devices is advisable—but will more time be necessary for individuals with different abilities? A longer transition window, with a two-stage transition, may help the driver to break concentration from media and redirect cognitive resources to driving and possibly increase safety during and after the transition, which will occur in challenging areas.

Additionally, if a driver is using a handheld device, will it be necessary for the vehicle's systems to interact with this device, presenting messages on the screen the driver is looking at, in addition to presenting messages on the vehicle's displays?

Study Method and design

Equipment

This study used the immersive driving simulator provided by Realtime Technologies at Stanford University, comprised of a complete sedan surrounded by a 270° seamless curved front screen and rear screen to provide an all-around immersive experience (see Figure 1). The simulator features an automated driving capability, allowing for switched control between the participant and the ADAS (Advanced Driver Assistance System). This allows research on human reactions to automated driving and mode switching, in a controlled environment, without risk to participants, and without regard to actual ADAS limitations and affordances.

Participants were provided with a tablet (an iPad Mini) running a custom media application developed by the research team, which allowed participants to select a movie segment to watch and an e-book to read during the parts of the drive when the vehicle was driving under automated control. Three films and three readings were available for participants to choose from. These selections were chosen to appeal to a wide audience, especially as this study included an age range of 15-81). This application also presented on-tablet warnings for the pre-alert and takeover messages superimposed over the movie or reading (see Figure 2).

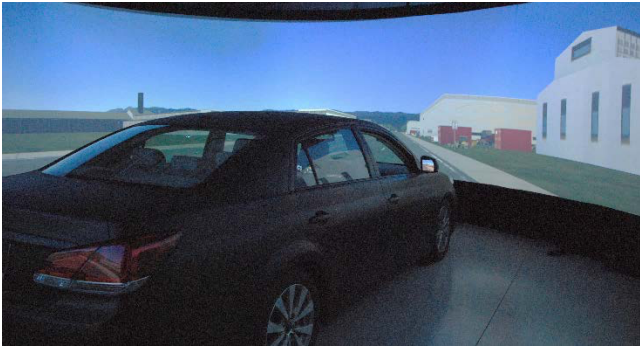


Figure 1. The driving simulator offers a high degree of presence, eliciting realistic responses to simulated threats.

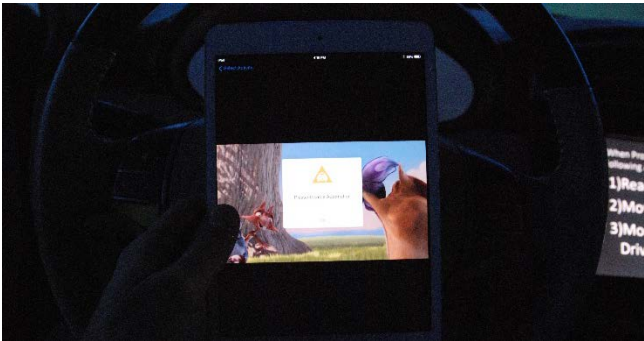


Figure 2. Alert messages were presented on the tablet, superimposed over the video or e-book.

Experimental Design

This study employed a 3x3x2 mixed design (task x location of information x handoff stages). Study participants drove in simulation for approximately 40 minutes, with three structured transitions from automated control to full driver control, shortly ahead of a potential hazard area such as a construction zone or school zone. This was intended to test reactions in areas where a structured handoff would be likely to occur in a near-future automated vehicle scenario (NHTSA level 3).

During the automated driving segments, participants performed one of three tasks: reading a selection from a book, watching a selection of a movie, and supervising the ADAS (see Table 1). To address ordering effects, the order of the tasks was varied between participants using a partial Latin Square design.

	Message on Instrument Panel	Messages on Instrument Panel and Tablet
Single Stage Handoff	<u>Nested Activities</u>	
Two-Stage Handoff	Reading	Video
	Supervision	

Table 1. Experimental Conditions

Participants

Three groups of twelve participants were recruited from local high schools, the student population at Stanford University, and from local senior centers. Participants ranged from 15 to 81 years of age, and

were segmented into three groups: young drivers (high school students 15-18 years old, $M=16.8$, $SD=.92$; 8 male, 4 female), college students and adults (17-69 years old, $M=28.4$, $SD=13.21$; 3 male, 9 female) and seniors (70-81 years old, $M=73.3$, $SD=5.17$; 6 male, 6 female). The groups were selected to assess whether inexperienced and young drivers, and seniors whose faculties had begun to degrade would react differently to task switching in driving.

Procedure

Before coming in to the lab, participants were given an online survey to complete at home, which included demographic questions, the Media Multitasking Inventory (Ophir, Nass, and Wagner 2009), a set of questions to assess driver sensation-seeking behavior (Yagil 2001), and the Multidimensional Traffic Locus of Control scale (Özkan and Lajunen 2005). These instruments were used to investigate the profiles of the different age groups, in order to better understand the relationship between driver traits and their behavior in the study.

Upon arrival, participants were given a pre-drive questionnaire inquiring about their current mood, using the International Positive and Negative Affect Schedule (PANAS) Short version (Watson, Clark, and Tellegen 1988), and their level of sleep debt using the Cleveland Adolescent Sleep Questionnaire (Spilsbury et al. 2007).

Once in the simulator, participants were briefed on the outline of the experiment, with detailed instructions on how the simulated vehicle would operate, how to activate and deactivate the automated driving system, and on the activities to perform while the car was in automated driving mode. Participants were given a tablet (an iPad Mini) for the media tasks (watching a movie and reading a book), and a clipboard with a set of surveys to be taken during the study. After participants told the experimenter that they fully understood the protocol, the in-vehicle recording system was started and the experiment begun.

Each automated driving section was 8.5 minutes long, intended to provide enough time for the driver to become bored with overseeing the automated driving system (and to begin to experience symptoms of sleepiness), or to fully engage in reading or watching the movie, before being presented with an alert to retake control of the vehicle ahead of a potentially hazardous area, such as a construction site or school zone (see Figure 3).

Events

Participants experienced four critical events: two car cut-off events and two pedestrian incursions. The first of the two car cut-off events occurred before any of the automated driving experiences, allowing for its use as a baseline event, in order to determine differences in reactions between groups. The second car cut-off and the two pedestrian incursions each occurred 1.5 seconds after the transition from automated to manual control, in order to assess whether drivers can re-engage in control of the vehicle and then avoid a collision shortly after taking control. After each event, the participant continued to drive under manual control for 30 seconds before re-enabling the automated driving system and entering the next block (see Figure 3).

The pedestrian incursion event was triggered according to the driver's speed - the pedestrian would walk into the road to cause a collision at 1.5s after the driver crossed the triggering point, if the driver

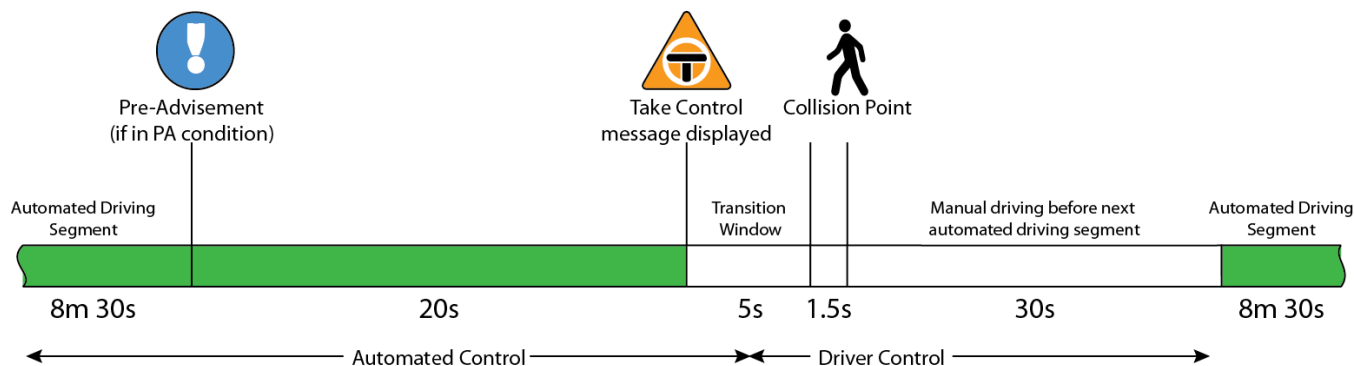


Figure 3. Outline of the structured transition and critical event sequence.

continued at constant speed. If the driver slowed or accelerated after the trigger point, the pedestrian would pass ahead of or behind the participant's vehicle.

The car cut-off event, where a vehicle would pass the participant's vehicle and then swerve into the participant's lane, fixed the time-to-collision at 1.5s by matching the speed of the driver's vehicle and then swerving into the lane and decelerating at a rate requiring a braking or steering response in order to avoid a collision. Reaction time and gap distance (Green 2013) was measured by the simulator and used as a measure of performance.

Structured Transitions

Each switch from driver control to computer control, or from computer control to driver control, was signaled by a voice command and visual indication on the instrument panel, with the visual indicator counting up or down to indicate transition time remaining (within a five-second window, see Figure 3) before the automated system would be automatically enabled or disabled. A button on the steering wheel was used to enable or disable the ADAS. If the driver did not press the button to switch modes within the five-second window, the ADAS would release control at the end of the window, giving drivers full control. Automation state was indicated by a visual indicator in the center of the instrument panel (see Figure 4).

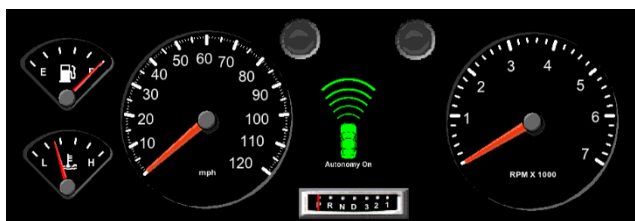


Figure 4. The automated driving system state is indicated by the icon in the center; state changes are depicted by an increasing or decreasing number of arcs above the car icon.

A pre-cue message was presented on the instrument panel to participants in the pre-cue present conditions, 20 seconds before the start of the handoff, (see Figure 5). For participants in the conditions where information was presented on both tablet and instrument panel, identical symbols were shown on the tablet and instrument panel simultaneously (see Figure 2).



Figure 5. The pre-alert, shown on the right side of the instrument panel, showed the driver that she or he was approaching an area where a handoff was likely to occur.

Media and Supervision Tasks

Shortly after enabling the automated driving system in each test block, participants were instructed by a computer-generated voice to fill out a paper survey, which included a question on their current state of alertness (the Stanford Sleepiness Scale (Dement 2010)), and a selected set of questions from the International PANAS-Short Form (Watson, Clark, and Tellegen 1988) to assess reactions to the prior transfer of control from the ADAS to the driver.

After completing the survey, each participant was instructed to either supervise the ADAS, watch a movie, or read a book section. The movie and readings were presented on a tablet, to evaluate driver interactions with the brought-in device. For participants in the message-on-tablet conditions, visual messages were presented on the tablet, as well as on the instrument panel (see Figure 6). The pre-advisement message would be presented above the video window, the take control message was presented overlaid over the video, which would continue to play under the message, and if the message was dismissed. The application would automatically reset to the activity selection page one minute after the event.

A post-drive questionnaire was administered after participants finished the driving component of the study, assessing subjective reactions to the structured transitions and critical events.

Results

Differences Between Age Groups

There are significant differences between the age-group cohorts in terms of their self-reported driving styles, which may explain some of the variance between groups in terms of evasion strategies. The Driving Style Inventory (Taubman-Ben-Ari, Mikulincer, and Gillath 2004) showed significant differences between groups on two

dimensions: risky driving ($F(2,33)= 5.509, p<.01$), and patient driving ($F(2,32)= 3.540, p<.05$). The DSI measure for anxiety was near-significant, $F(2,33)= 3.117, p=.058$. Note: the differences in degrees of freedom is due to participants not answering all questions in the battery. Post-hoc testing using the Tukey HSD for multiple comparisons explained the differences as being driven differently in each category. On the anxiety dimension, the adult cohort ($M= 11.911, SD=3.088$) had a significantly higher mean than the young ($M=8.917, SD=2.065$) and old ($M=9.250, SD=2.734$) cohorts, $p<.05$. On the dimension of risky driving, the young drivers rated themselves as significantly more risky drivers ($M=16.167, SD=10.659$) than adults ($M=9.45, SD=4.887$), $p<.05$ and older drivers ($M=7.750, SD=1.357$), $p<.05$. Adults and seniors did not differ significantly from each other on this measure. On the dimension of patience, older drivers rated themselves significantly higher ($M=19.583, SD=2.275$) than adults ($M=15.182, SD=4.490$), $p<.05$; and young drivers ($M=17.333, SD=4.716, p<.05$, see Table 2 and Figure 6.

		N	Mean	Std. Deviation
DSI: Anxious	New Drivers	12	8.917	2.065
	Adults	12	11.417	3.088
	Seniors	12	9.250	2.734
DSI: Risky	New Drivers	12	16.167	10.659
	Adults	12	8.667	4.887
	Seniors	12	7.750	1.357
DSI: Patient	New Drivers	12	17.333	4.716
	Adults	11	15.182	4.490
	Seniors	12	19.583	2.275

Table 2. Driving Style Inventory: self-reported measures for anxiety, risky driving, and patient driving

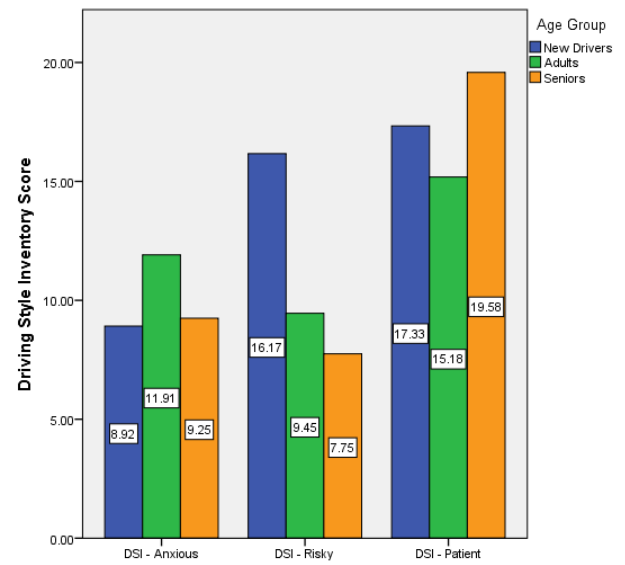


Figure 6. Driving Style Inventory: self-reported measures for anxiety, risky driving, and patient driving

The difference in risk-seeking/risk aversion may explain in part the behavioral differences between groups, as older drivers adjust their behavior in order to avoid situations potentially beyond their abilities (Nishida 1999). If older drivers are more patient and more risk-averse than the other groups, then their behavior upon release from automated driving can be put in context, as they tend to slow upon release from automated driving, as a result of coasting down before accelerating again. This may be due to missing the audible cue, and failure to realize that the ADAS had released control to the driver until the vehicle speed slowed appreciably.

There was not a significant difference in maximum or mean speeds between age cohorts for the baseline event, before any of the automated driving segments, indicating the risk-seeking behavior shown in the first post-transition event was not indicative of a global trend of speeding. Minimum speed during the baseline event was significantly different between groups, the older drivers slowing to a lower speed ($M=11.356$ m/s, $SD=7.96$) than the other two groups (adults $M=17.333$ m/s, $SD=6.999$; young drivers $M=18.203$ m/s, $SD=6.131$), $F(2,33)=3.336, p<.05$. A post-hoc pairwise comparison using Tukey's HSD revealed a marginally significant difference between the young drivers and old drivers, $p=.060$.

Speed After Transition From Automated Driving

Post transition, drivers' behavior differed significantly in terms of mean ($F(2,33)= 3.956, p<.05$) and minimum speed ($F(2,33)= 6.541, p<.01$), with young drivers showing greater mean and minimum speeds in the post-transition period, after the first automated driving section. Multiple comparisons using Tukey's HSD showed a significant difference between the young drivers and the adults, and between the young drivers and seniors.

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($p < .05$), and a marginally significant ($p = .062$) difference between the young drivers and seniors, see Table 3.

		N	Mean	Std. Deviation
First Event Mean Speed (m/s)	New Drivers	12	24.876	2.485
	Adults	12	20.104	7.446
	Seniors	12	20.386	1.896
	Total	36	21.789	5.041
First Event Minimum Speed (m/s)	New Drivers	12	20.799	2.922
	Adults	12	15.377	7.096
	Seniors	12	14.271	2.876
	Total	36	16.816	5.429

Table 3. First post-transition event (car cut-off) mean and minimum speed, by age group.

Evasion Strategies

The different age cohorts employed different evasion strategies: young and adult drivers used primarily braking evasion strategies, and older drivers used combination braking and steering evasion strategies in the first car-cut off event, $\chi^2(4) = 21.72$, $p < .01$; see Table 4. The evasion strategy difference may be a result of different training and experience between groups, but it is impossible to know about evasion strategy choice without post-interviews. There was not a significant difference in maximal braking forces between groups, ruling out the possibility that the older drivers chose an evasion strategy based on potential braking ability.

		Age Group		
		New Drivers	Adults	Seniors
First Event Response (car-cut off)	braking and steering	1	0	9
	braking	11	11	3
	no response	0	1	0
	Total	12	12	12

Table 4. First event (car cut-off, before any automated driving segment) response strategy, by age group.

The first of the two pedestrian incursions also showed differences between age cohorts: seniors slowed more when the automated driving system released control, and as a result of a lower speed were more frequently able to avoid the pedestrian without a braking or steering response.

		Age Group		
		New Drivers	Adults	Seniors
Third Event Response (pedestrian incursion)	braking and steering	1	2	3
	braking	10	8	1
	steering	1	1	0
	no response	0	1	8
Total		12	12	12

Table 5. Third post-transition event (pedestrian incursion) response strategy, by age group.

Discussion

Collision Avoidance

Four critical events were placed along the course to test driver reactions, one before any of the media or supervision segments, and one after each media or supervision task. There was not a significant difference in collisions, when compared across age groups or handoff staging/information location, for any of the events. This indicates that the variations in speed and evasion initiation are sufficient to equalize the outcomes between groups.

That said, 38.9% of participants hit the first of two pedestrians, and 36.1% hit the second pedestrian, indicating that the surprise of a pedestrian incursion shortly after transition was likely to cause a collision. This indicates that a surprise shortly after a structured transition presents a safety hazard if the automated safety systems of the vehicle (automated collision avoidance) does not take control and avoid the collision. The two car cut-off events did not show a large number of crashes, there were only four collisions, all during the event before any automated driving segment.

Considering the large number of collisions with pedestrians shortly after the transitions, this reveals the necessity for safety systems to exert agency in such situations as drivers' abilities may be impaired, although not necessarily by switching from media to driving. This may result in a transfer of control from the human driver to the computer in a situation where an automated driving system would have difficulty (such as the accident area, construction zone, and school zone used in this study) and the computer may have to rapidly take back control in case the driver does not or cannot act to resolve a fast-unfolding critical situation such as a pedestrian incursion or car cut-off event.

Participants who delayed accelerating back to speed after the automated driving system was disabled were able to avoid the pedestrian without an evasive action. Eight of the seniors did not make an evasive action in the first of the two pedestrian incursions, but as their collision rate was not higher than the other age groups, the strategy was obviously effective. This behavior highlights a difference between older drivers and the younger drivers in that the older drivers would let the car coast for longer, reengaging with driving more slowly, unintentionally exploiting an artifact of the simulation design.

Challenges for Older Drivers

A large fraction of the seniors failed to follow the directions, and had to be dropped from the analysis and replaced with others in the same conditions. Thirty percent of seniors failed to follow the protocol (7/22), a much higher fraction than the young and adult drivers. That such a large number of seniors had difficulty following directions given minutes before does not bode well for the need for drivers to recall and use complex information about automated driving systems long after receiving instruction from the dealer or reading the manual, if indeed they do get any instruction at all.

Memory and cognitive processing declines among the elderly present a challenge for highly automated vehicle design, when drivers are required to take control of the vehicle (NHTSA levels 2 and 3). While there is enormous diversity among seniors, even at the same age, the hoped-for benefits may be more difficult to attain than some would desire. That over 1/3 of seniors had to be rejected due to failure to follow the experimental protocol indicates that there will be an enormous challenge to design vehicle systems that can cope with the issues presented by drivers suffering from age related impairments. Especially in light of the hopes many have for automation to increase the ability for the elderly to travel, it will be some time until systems are good enough to cope with drivers who have significant difficulty in formulating a response in a situation where the driver is forced to react quickly to avoid a collision. Even with collision avoidance abilities, inappropriate driver actions may make it more difficult or impossible for such systems to achieve a 'save' by braking or initiating an evasive maneuver.

These concerns stem from analysis of the driving of several participants who failed to follow the protocol, but who drove through the entire experiment and experienced the full set of four events. These participants illustrated the challenges presented by the elderly who have age-related declines in driving ability. Elderly drivers tune their behavior to compensate for decreased abilities by driving more slowly (Nishida 1999) and more cautiously. Automated driving systems will follow the speed limits and rules of the road, and thus may operate in ways that make seniors uncomfortable, or release seniors into driving situations that are very challenging for them—e.g. relinquishing control at a higher speed than the driver is used to, or in a difficult driving situation. These are design considerations that need to be addressed for the success and safety of automation specifically geared towards the elderly.

Challenges for novice drivers

The young drivers exhibit a greater orientation towards risk-seeking, and indeed do have higher average and minimum speeds during the critical event situations, but did not have more collisions or shorter minimum gap during the car cut-off events. The fast reactions of the younger drivers compensated to some degree for their riskier behavior, allowing them to still drive safely by objective standards. As these situations (a car cutting the driver off, a pedestrian incursion) have clear avoidance behaviors, they are not particularly challenging from the standpoint of a novice driver. More challenging situations such as icy roads or complex traffic evaluation may present greater problems for novices, and thus yield a differential outcome between experience levels.

If young and less experienced drivers use automation a significant amount of the time, they may not acquire or practice the skills necessary for safe driving in adverse circumstances, thus becoming

more dangerous in difficult situations. This de-skilling may occur for drivers of all ages as well. This will present another design challenge for automated driving—keeping the driver trained in order to be able to drive safely when called on. It may be possible to design automated systems to teach drivers to maintain and improve their skills, countering the potential de-skilling extensive reliance on automation may cause.

Handoff Staging

That there was not a significant difference between the single stage transfer of control to the driver and the two-stage transfer with the pre-advisement indicates that the five-second structured handoff is sufficient time for drivers to take control of the vehicle and that the pre-advisement did not provide any advantage post-transition in terms of collision avoidance. Because the transitions occurred on straight road segments, for the most part drivers had a relatively easy time resuming driving, and then were surprised by the critical event.

In real-world situations, only some transition events can be forecast 20 seconds ahead, and the situations used in the study (an accident scene with emergency services present, a construction zone, and a school zone) are situations where the car could be aware ahead of time. If with reasonably alert drivers (the participants in this study were engaged in media consumption or supervising the ADAS, and none were obviously asleep for any prolonged length of time), structured handoffs can be relatively short. The five-second transitions of control in this study, which all occurred on straight roads, were long enough for many drivers to safely re-engage in driving, but this may not be the case in a shorter transition or one which releases the driver into a more challenging situation such as in a curve or onto an icy road. If drivers are to sleep (setting aside concerns about occupant protection), more time may be necessary for arousing a driver, or the automated system may need to act independently if the driver cannot be engaged in time. One possible solution is the development of collaborative control systems that can dynamically share control between the computer and driver (Miller and Ju 2015).

Location and Modality of Transition Messages

The presentation of takeover (and for participants in the pre-advisement conditions, the pre-advisement messages) on the tablet as well as on the instrument panel did not significantly change post-transition collision avoidance ability. This is likely due to audible cue presented at the same time, and the length of the transition window. There were several seniors with hearing impairments, and two who did not wear their hearing aids had difficulty hearing the simulator's voice commands and were subsequently dropped from the manipulation analysis as a result of not following the protocol. If vehicles are to communicate with audible alerts, including speech, systems must be adaptive to accommodate those with sensory impairments.

Conclusions

To address the needs of a diverse driving population, automated driving systems need to take in to account the abilities of the driver in terms of how to transfer or share control, and active safety systems such as collision avoidance need to intervene in the case where a hazardous situation presents itself shortly after transition of control. Even though a large fraction of participants struck the pedestrians, there was not a correlation with age group or media use as an

antecedent, which implies that the surprising nature of the event played a large role in terms of participants' failure to avoid the pedestrians.

Considering the behavior of participants who failed to follow the protocol, they did have more collisions as a fraction than participants who were able to follow the protocol, likely as a result of having superior physical and mental condition. If drivers who have hearing impairments or cognitive decline are more likely to have collisions, automation must be designed to provide extra support.

Media consumption will inevitably be part of driving in the future, once automation makes it possible to safely hand over driving for extended periods to the ADAS. Communicating with brought-in devices to present messages does not appear to give an advantage in handoff situations, but this may not be the case for hearing impaired individuals who have difficulty in hearing audible alerts from the vehicle which would divert attention from media interaction. Additionally, if a media device is used with headphones or at high volume, and is not in communication with the vehicle, audible alerts may be masked—potentially causing a hazardous situation.

Care needs to be taken with regards to designing automated systems for driving assistance, specifically taking into account the abilities of drivers to react to state changes and to act appropriately. As the time between any training on system operation at vehicle purchase and a situation where the system requires human intervention—and a critical event presents itself at the same time may be long, it will be hard for drivers (especially seniors) to deal with the challenges.

Previous research (e.g. the HAVEit project (Rauch et al. 2009)) has discussed the need for driver state assessment, which will be necessary in models where the driver will be required to engage in driving (e.g. NHTSA levels 2 and 3). Future research will need to investigate how drivers and automated systems can interact to avoid collisions when in a collaborative control regime. With a diverse population of drivers, it will be necessary not only to know the driver's current state, but also the driver's potential ability to control the vehicle under adverse conditions, if the driver is expected to be able to control the vehicle in a situation where the automated system cannot.

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